

University of Science and Technology Houari Boumediene

Faculty of Computer Science



NETWORKS 2

TCP/IP Networks: Architecture and Operation

Chapter 2: Physical and Logical Addressing

Teaching Unit: UEF61

Credits: 4 | Coefficient: 3

Weekly: 1h30 Lecture + 1h30 Lab

Target Audience: Third-Year Undergraduate Students

Prerequisites: Computer Architecture, Networks 1 (Introductory)

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Table of Contents

Table of Contents.....	2
Chapter 2: Physical and Logical Addressing.....	4
Learning Outcomes.....	4
1. Physical Addressing (MAC Addresses).....	4
1.1 Introduction to MAC Addresses.....	4
1.2 MAC Address Structure.....	4
1.3 Types of MAC Addresses.....	5
Unicast Address.....	5
Multicast Address.....	5
Broadcast Address.....	5
1.4 MAC Address Table.....	6
2. IPv4 Logical Addressing.....	6
2.1 IPv4 Address Structure.....	6
2.2 IPv4 Address Classes (Classful Addressing).....	7
2.3 Special IPv4 Addresses.....	7
2.4 Public vs Private IP Addresses.....	8
3. Subnetting.....	8
3.1 Introduction to Subnetting.....	8
3.2 Subnet Masks.....	8
3.3 CIDR Notation.....	9
3.4 Subnetting Calculations - Step by Step.....	9
3.5 VLSM (Variable Length Subnet Masking).....	10
4. Introduction to IPv6 Addressing.....	11
4.1 IPv6 Motivation.....	11

4.2 IPv6 Address Format.....	11
4.3 IPv6 Address Types.....	11
4.4 IPv6 Address Compression Rules.....	12
5. Lab Activities.....	12
Lab 2.1: MAC Address Analysis.....	12
Lab 2.2: Subnetting Practice.....	13
6. Chapter Summary.....	13
7. Review Questions.....	14
Conceptual Questions.....	14
Problem-Solving Questions.....	14
8. Exercises.....	15
Bibliography.....	16
Online Resources.....	16

Chapter 2: Physical and Logical Addressing

Learning Outcomes

After completing this chapter, students will be able to:

1. Explain MAC addressing, including unicast, multicast, and broadcast addresses;
2. Design and implement IPv4 addressing schemes using classful and classless approaches;
3. Perform subnetting calculations including subnet masks, CIDR notation, and VLSM;
4. Distinguish between public and private IP addresses and understand NAT concepts;
5. Describe IPv6 addressing structure and representation.

1. Physical Addressing (MAC Addresses)

1.1 Introduction to MAC Addresses

Media Access Control (MAC) addresses are hardware addresses that uniquely identify network interfaces at the data link layer. Each network interface card (NIC) is assigned a unique 48-bit MAC address by its manufacturer.

Definition: MAC Address -A 48-bit hardware address burned into the network interface card (NIC) during manufacturing. Expressed as six hexadecimal bytes separated by colons or hyphens, (the mac address is represented by numbers and letters from A to F). (e.g., 00:1A:2B:3C:4D:5E).

1.2 MAC Address Structure

The 48-bit MAC address is divided into two parts:

- Organizationally Unique Identifier (OUI): The first 24 bits (3 octets) identify the manufacturer of the NIC.

- **Device Identifier:** The last 24 bits (3 octets) are unique number from the manufacturer that identifies each device on a network (identify the specific device).

Example MAC Address Breakdown:

MAC Address: 00:1A:2B:3C:4D:5E
OUI (Manufacturer): 00:1A:2B (Cisco Systems, Inc.)
Device ID: 3C:4D:5E (Unique to this NIC)

1.3 Types of MAC Addresses

Unicast Address

A unicast address identifies a single network interface. When a frame is sent to a unicast address, only the intended recipient processes it. The least significant bit of the first octet is 0 for unicast addresses.

Example: Unicast address

00:1A:2B:3C:4D:5E - First octet (00) in binary: 00000000. LSB is 0, indicating unicast.

Multicast Address

A multicast address identifies a group of network interfaces. Frames sent to multicast addresses are processed by all members of the group. The least significant bit of the first octet is 1 for multicast addresses.

Example: Multicast address

01:00:5E:00:00:01 - First octet (01) in binary: 00000001. LSB is 1, indicating multicast. This specific address is used for IPv4 multicast to all systems on the local subnet.

Broadcast Address

The broadcast address (FF:FF:FF:FF:FF:FF) is a special multicast address that

indicates all devices on the local network segment. Every device must process frames sent to this address.

Example: Broadcast Usage

When a device needs to discover the MAC address associated with an IP address (ARP request), it sends the request to the broadcast address so all devices on the segment receive it.

1.4 MAC Address Table

Switches maintain MAC address tables (also called CAM tables) to forward frames efficiently (see table 2.1).

MAC Address	Port	VLAN	Type
00:1A:2B:3C:4D:5E	Gi0/1	1	Dynamic
00:1A:2B:3C:4D:5F	Gi0/2	1	Dynamic
00:1A:2B:3C:4D:60	Gi0/3	10	Static

Table 2.1: Mac table example

2. IPv4 Logical Addressing

2.1 IPv4 Address Structure

An IPv4 address is a 32-bit number that uniquely identifies a device on an IP network. It is typically written in dotted-decimal notation as four decimal numbers (octets) separated by periods.

IPv4 Address Representation:

Binary: 11000000.10101000.00000001.00000001

Decimal: 192.168.1.1

Hexadecimal: C0.A8.01.01

2.2 IPv4 Address Classes (Classful Addressing)

Originally, IPv4 addresses were divided into five classes (A, B, C, D, E) based on the first few bits of the address:

Class	First Bits	First Octet Range	Default Mask	Networks	Hosts/Network
A	0	1-126	255.0.0.0 (/8)	126	16,777,214
B	10	128-191	255.255.0.0 (/16)	16,384	65,534
C	110	192-223	255.255.255.0 (/24)	2,097,152	254
D	1110	224-239	N/A	Multicast	N/A
E	1111	240-255	N/A	Reserved	N/A

Table 2.2: IPv4 Address Classes

2.3 Special IPv4 Addresses

Address/Range	Purpose	Description
0.0.0.0/8	Current network	Source address for bootstrapping
127.0.0.0/8	Loopback	Local host testing (127.0.0.1)
255.255.255.255	Limited broadcast	Broadcast to local network
10.0.0.0/8	Private Class A	RFC 1918 private network
172.16.0.0/12	Private Class B	RFC 1918 private network
192.168.0.0/16	Private Class C	RFC 1918 private network
169.254.0.0/16	Link-local	APIPA auto-configuration

Table 2.3: Special IPv4 Addresses

2.4 Public vs Private IP Addresses

RFC 1918 defines three ranges of private IP addresses that are not routable on the public Internet:

- 10.0.0.0/8 (10.0.0.0 to 10.255.255.255) - Single Class A network
- 172.16.0.0/12 (172.16.0.0 to 172.31.255.255) - 16 Class B networks
- 192.168.0.0/16 (192.168.0.0 to 192.168.255.255) - 256 Class C networks

Private addresses require Network Address Translation (NAT) to communicate with the public Internet. This design conserves the limited IPv4 address space.

3. Subnetting

3.1 Introduction to Subnetting

Subnetting is the process of dividing a network into smaller subnetworks (subnets). This improves network efficiency, enhances security, and allows more flexible address allocation.

Definition: Subnet -A logical subdivision of an IP network. Subnetting involves borrowing bits from the host portion of an address to create a subnet identifier.

3.2 Subnet Masks

A subnet mask is a 32-bit value that distinguishes the network/subnet portion from the host portion of an IP address. In binary, network bits are represented by 1s and host bits by 0s.

Subnet Mask Examples:

Class A: 255.0.0.0 = 11111111.00000000.00000000.00000000

Class B: 255.255.0.0 = 11111111.11111111.00000000.00000000

Class C: 255.255.255.0 = 11111111.11111111.11111111.00000000

3.3 CIDR Notation

Classless Inter-Domain Routing (CIDR) notation expresses the subnet mask as a slash followed by the number of network bits. This compact format has largely replaced classful addressing.

CIDR	Subnet Mask	Binary Mask	Total Hosts
/8	255.0.0.0	11111111.00000000.00000000.00000000	16,777,216
/16	255.255.0.0	11111111.11111111.00000000.00000000	65,536
/24	255.255.255.0	11111111.11111111.11111111.00000000	256
/25	255.255.255.128	11111111.11111111.11111111.10000000	128
/26	255.255.255.192	11111111.11111111.11111111.11000000	64
/27	255.255.255.224	11111111.11111111.11111111.11100000	32
/28	255.255.255.240	11111111.11111111.11111111.11110000	16
/30	255.255.255.252	11111111.11111111.11111111.11111100	4

Table 2.4: CIDR Notation

3.4 Subnetting Calculations - Step by Step

The following example demonstrates subnetting a Class C network:

Example: Subnetting 192.168.1.0/24 into 4 subnets

Step 1: Determine number of subnet bits needed

- Need 4 subnets
- Formula: $2^n \geq \text{required subnets}$
- $2^2 = 4$, so need 2 subnet bits

Step 2: Calculate new subnet mask

- Original: /24 (255.255.255.0)
- New: $/24 + 2 = /26$

- New mask: 255.255.255.192

Step 3: Determine subnet size

- Host bits remaining: $32 - 26 = 6$

- Hosts per subnet: $2^6 - 2 = 62$

Step 4: List subnet ranges

- Subnet 1: 192.168.1.0/26 (0-63)

- Subnet 2: 192.168.1.64/26 (64-127)

- Subnet 3: 192.168.1.128/26 (128-191)

- Subnet 4: 192.168.1.192/26 (192-255)

Step 5: Identify usable host addresses

- Subnet 1: 192.168.1.1 to 192.168.1.62

- Network: 192.168.1.0, Broadcast: 192.168.1.63

3.5 VLSM (Variable Length Subnet Masking)

VLSM allows subnets of different sizes within the same network, optimizing address allocation. This is essential for efficient IP address management.

VLSM Example: 192.168.10.0/24

Requirements:

- Subnet A: 50 hosts
- Subnet B: 25 hosts
- Subnet C: 10 hosts
- Subnet D: 2 hosts (point-to-point)

Solution:

1. Subnet A (/26): 192.168.10.0/26 - 62 usable hosts
2. Subnet B (/27): 192.168.10.64/27 - 30 usable hosts
3. Subnet C (/28): 192.168.10.96/28 - 14 usable hosts
4. Subnet D (/30): 192.168.10.112/30 - 2 usable hosts

4. Introduction to IPv6 Addressing

4.1 IPv6 Motivation

IPv6 was developed to address IPv4 address exhaustion and provide enhancements including:

- Vastly larger address space (128 bits vs 32 bits)
- Simplified header format for faster processing
- Built-in security features (IPsec)
- Better support for Quality of Service (QoS)
- Stateless auto-configuration

4.2 IPv6 Address Format

IPv6 addresses are 128 bits, represented as eight groups of four hexadecimal digits separated by colons.

IPv6 Address Examples:

Full: 2001:0DB8:0000:0000:0000:FF00:0042:8329

Compressed: 2001:DB8::FF00:42:8329

Loopback: ::1

Unspecified: ::

4.3 IPv6 Address Types

Type	Prefix	Description
Unicast	Various	Identifies a single interface
Link-Local	FE80::/10	Auto-configured, non-routable
Unique Local	FC00::/7	Private addressing (like RFC 1918)
Global Unicast	2000::/3	Routable addresses
Multicast	FF00::/8	One-to-many communication

Type	Prefix	Description
Anycast	Any unicast	One-to-nearest communication

Table 2.4: IPv6 Address Types

4.4 IPv6 Address Compression Rules

IPv6 addresses can be compressed using these rules:

1. Leading zeros in each group can be omitted
2. One group of consecutive all-zero sections can be replaced with ::
3. The :: can only be used once per address

Example: Compression Examples

2001:0DB8:0000:0000:0000:0000:1428:57AB -> 2001:DB8::1428:57AB

FF01:0000:0000:0000:0000:0000:0000:0101 -> FF01::101

0000:0000:0000:0000:0000:0000:0000:0001 -> ::1

5. Lab Activities

Lab 2.1: MAC Address Analysis

Objective: Identify and analyze MAC addresses in a network environment.

Commands:

Windows: ipconfig /all

Linux: ip link show

macOS: ifconfig -a

Cisco: show mac address-table

Tasks:

1. Identify your computer's MAC address
2. Determine the OUI (manufacturer) using online databases
3. Observe MAC address table on a switch
4. Identify different types of MAC addresses (unicast, multicast, broadcast)

Lab 2.2: Subnetting Practice

Objective: Master subnetting calculations through practical exercises.

Exercise A - Basic Subnetting:

1. Given: 172.16.0.0/16, need 8 subnets
Find: New subnet mask, subnets, hosts per subnet
2. Given: 10.0.0.0/8, need 1000 subnets
Find: New subnet mask and first 5 subnet ranges
3. Given: 192.168.5.0/24, need subnets with 20 hosts each
Find: Appropriate subnet mask and subnet ranges

Exercise B - VLSM Design:

Network: 10.10.0.0/16
Requirements:
- HQ LAN: 500 hosts
- Branch1 LAN: 200 hosts
- Branch2 LAN: 100 hosts
- WAN Link 1: 2 hosts
- WAN Link 2: 2 hosts
Task: Design an efficient VLSM scheme.

6. Chapter Summary

This chapter covered the fundamental addressing schemes in TCP/IP networks:

- MAC addresses are 48-bit hardware addresses used at the link layer, with unicast, multicast, and broadcast types
- IPv4 addresses are 32-bit logical addresses, historically classful but now primarily classless (CIDR)
- Subnetting divides networks into smaller subnets using subnet masks

- VLSM enables efficient address allocation with variable-sized subnets
- Private addresses (RFC 1918) conserve public IPv4 space
- IPv6 provides 128-bit addresses with improved features and virtually unlimited address space.

7. Review Questions

Conceptual Questions

1. Why are MAC addresses necessary when IP addresses already uniquely identify devices?
2. Explain the difference between classful and classless addressing. Why did CIDR replace classful addressing?
3. What are the advantages and disadvantages of using private IP addresses?
4. Compare IPv4 and IPv6 addressing. What problems does IPv6 solve?

Problem-Solving Questions

1. Given network 172.20.0.0/16, create 16 subnets. List the network address, usable host range, and broadcast address for subnets 1, 8, and 16.
2. A company has been allocated 192.168.100.0/24. Design a VLSM scheme for: 3 departments with 50 hosts each, 4 departments with 25 hosts each, and 6 point-to-point WAN links.
3. Convert the following IPv6 addresses to compressed format:
2001:0DB8:0000:1234:0000:0000:0000:5678 and
FE80:0000:0000:0000:020C:29FF:FE12:3456

8. Exercises

Exercise 2.1: Design an IP addressing scheme for a fictional company with multiple offices. Document your VLSM design with a network diagram showing all subnets and address allocations.

Exercise 2.2: Research and write a 2-page report comparing IPv4 and IPv6 deployment status worldwide. Include statistics on IPv6 adoption and challenges in the transition process.

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